

CIE

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TITLE: A supervised machine learning framework for the detection of fraudulent blockchain accounts on Ethereum

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ABSTRACT

1.1 Problem Background

Over time, cryptocurrency scams has gone up a lot and is now a big financial concern. More than \$1 billion has been lost, affecting over 46,000 victims worldwide, which shows a major rise compared to 2018. Ethereum alone contributes to nearly 9% of such fraud cases. Although blockchain technology is known for security and transparency, its anonymous nature can sometimes create risks for users. Because of this, attackers to perform scams such as phishing, hacking, and fake transactions without being easily identified. Current fraud detection methods often struggle to manage the fast growth of transactions and the changing techniques used by attackers. So, a smarter and automated system is required now to detect fraud by analyzing transaction patterns more effectively.

1.2 Objective of the Project

The main objective of this project is to build a fraud detection system using supervised machine learning to correctly identify fraudulent Ethereum accounts. The project aims to achieve an F1-score above 90% and help crypto exchanges and wallet platforms.

1.3 Proposed Solution / System Overview

In this system, supervised machine learning is used to classify Ethereum transactions as fraudulent or non-fraudulent. It includes steps like collecting data, preparing it, choosing important features, training and testing the model. The dataset is cleaned by dealing with missing values, removing unnecessary and constant features, and converting category-type data into a usable format. The prepared data is then split into training and testing sets. Here, a Logistic Regression model is used to understand transaction behaviour and predict fraud. This approach keeps the system simple, efficient, and understandable fraud detection system.

1.4 Key Features or Functionality

The system tries to identify fraud patterns using transaction-based features like transaction frequency, time gaps, and token activity. It also includes cleaning the dataset, useful feature selection, categorical conversion, and data visualization. Correlation heatmaps help in understanding how features are related. The model also helps finding important factors that influence fraud detection and supports better understanding of suspicious Ethereum transaction behaviour.

1.5 Technology / Method Used

The project is built using Python. Pandas and NumPy help in handling and organizing data. While Seaborn and Matplotlib are mainly used for visualization. Scikit-learn helps in tasks such as data splitting, encoding, scaling, training, and testing the model. Logistic Regression is chosen as the main model since it is simple and effective for binary classification problems.

1.6 Expected Outcome & Benefits

The expected outcome is the accurate detection of fraudulent Ethereum transactions with a model accuracy of **99.14%** and a low error rate of **0.86%**. The model is also expected to achieve strong precision, recall, and F1-score values of around **98-99%**. The project also helps identify important fraud-related transaction patterns, helps improve blockchain security, and reduces the risk of financial loss for users and platforms.

1.7 Original Contribution Statement

This project shows a simple and effective machine learning pipeline for Ethereum fraud detection. It explains how data preparation, feature selection, transaction analysis, and Logistic Regression can be combined to create an accurate, understandable, and practical fraud detection system.